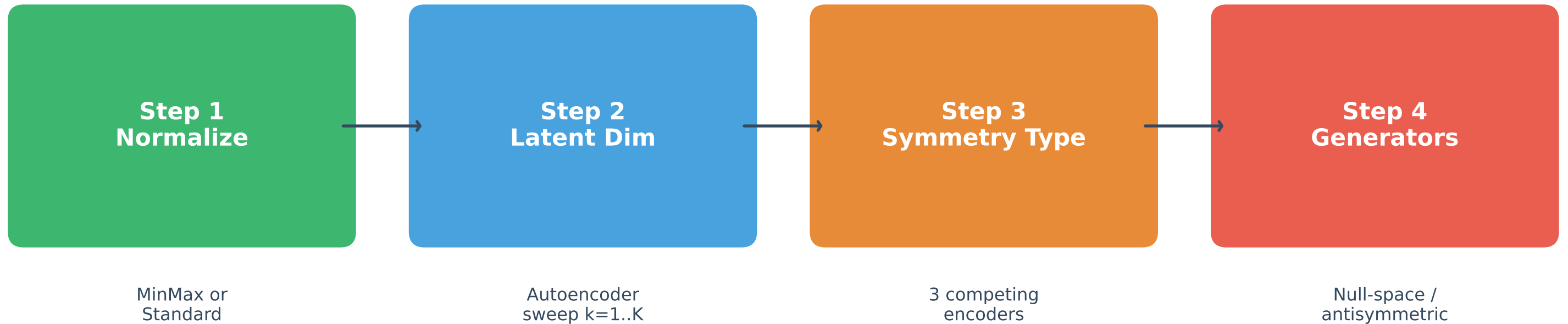


Symmetry Discovery in Physics Data

Stage1 Pipeline: From Raw Data to Lie-Algebra Generators

PyDimension Project

Pipeline Overview



Input: X (features), y (output) | Output: symmetry type + generators

Step 1: Data Normalization

MinMax Normalization

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Scales to [0, 1] — good for scaling symmetry

Standard Normalization

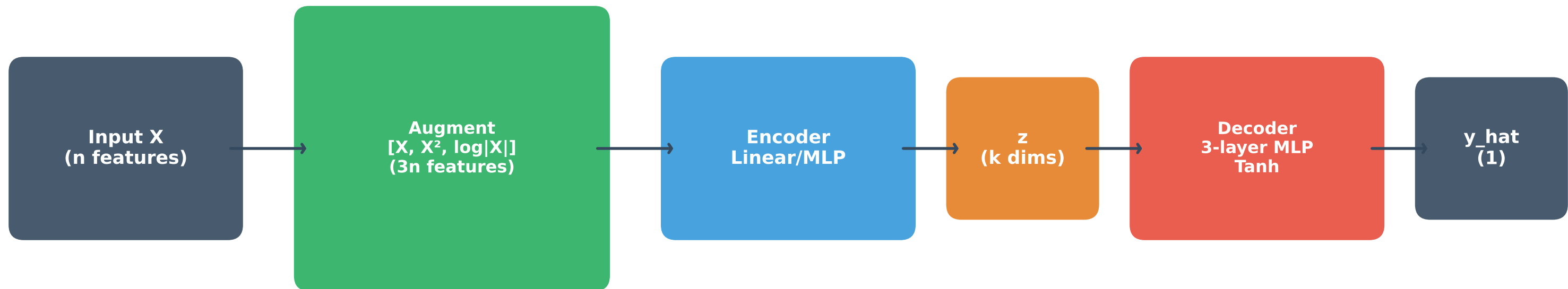
$$X_{norm} = \frac{X - \mu}{\sigma}$$

Zero mean, unit variance — good for translational

Choice of normalization affects which symmetry is easiest to detect

Positive data (e.g. physical quantities) works well with MinMax
Centered data (e.g. momenta) works well with Standard

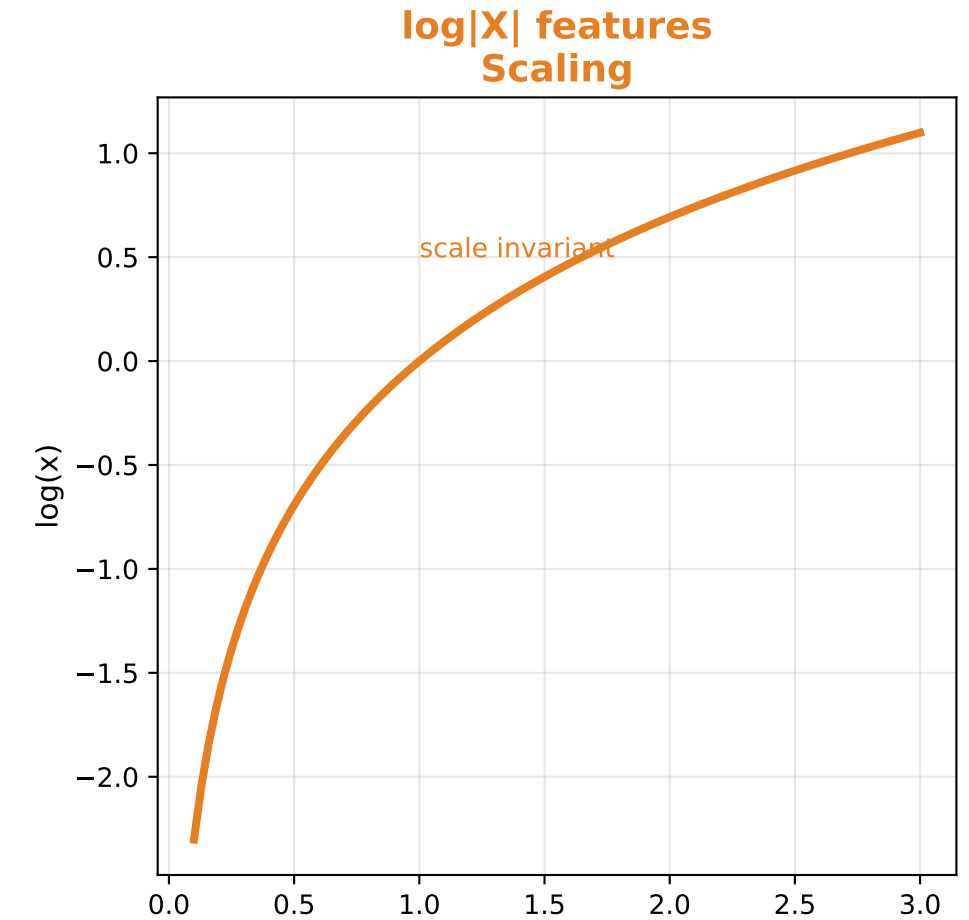
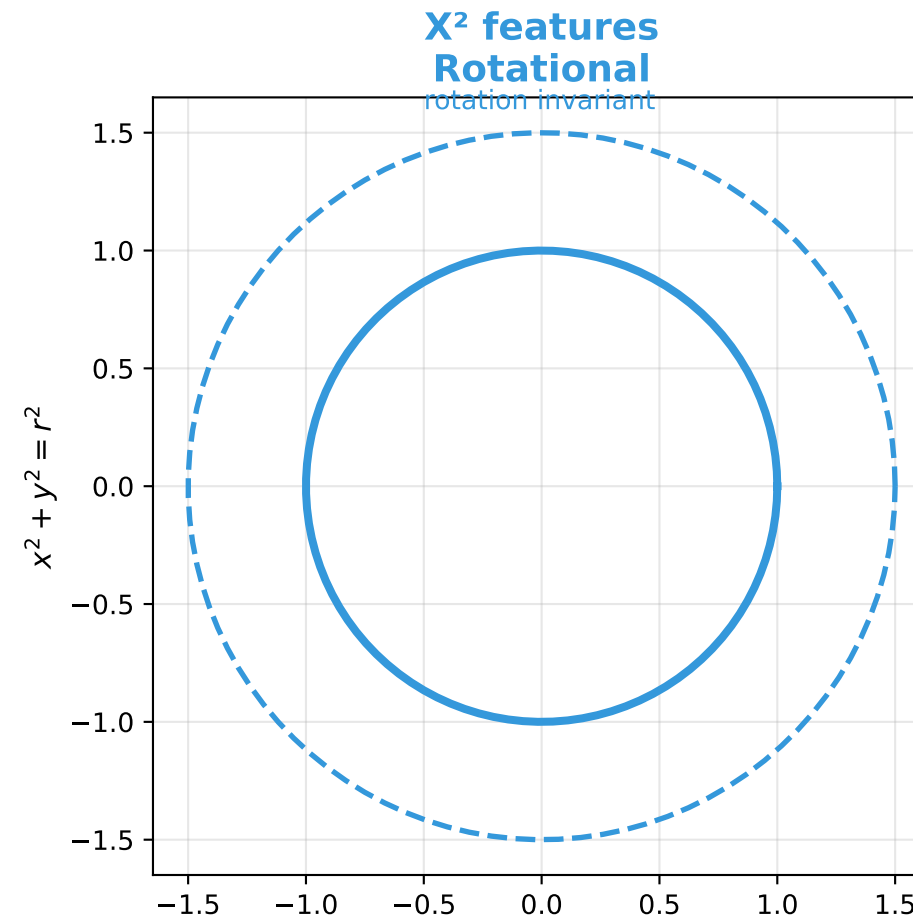
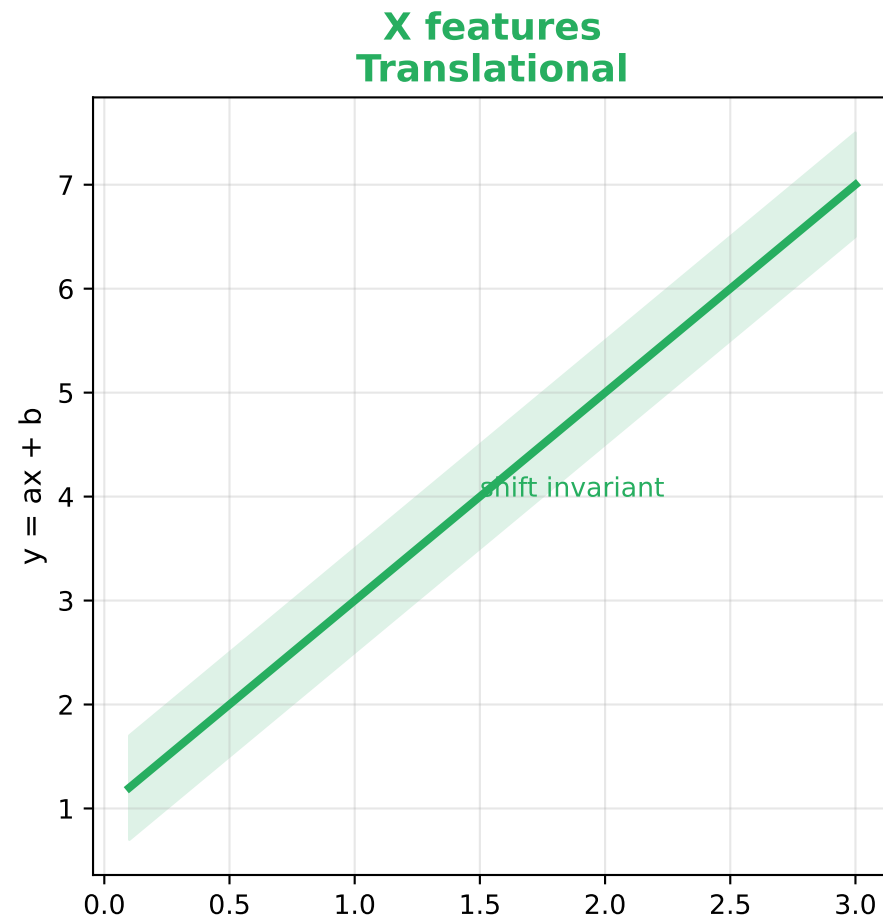
Step 2: Latent Dimension Discovery



Sweep $k = 1, 2, \dots, K$. Pick smallest k where $R^2 > \text{threshold}$

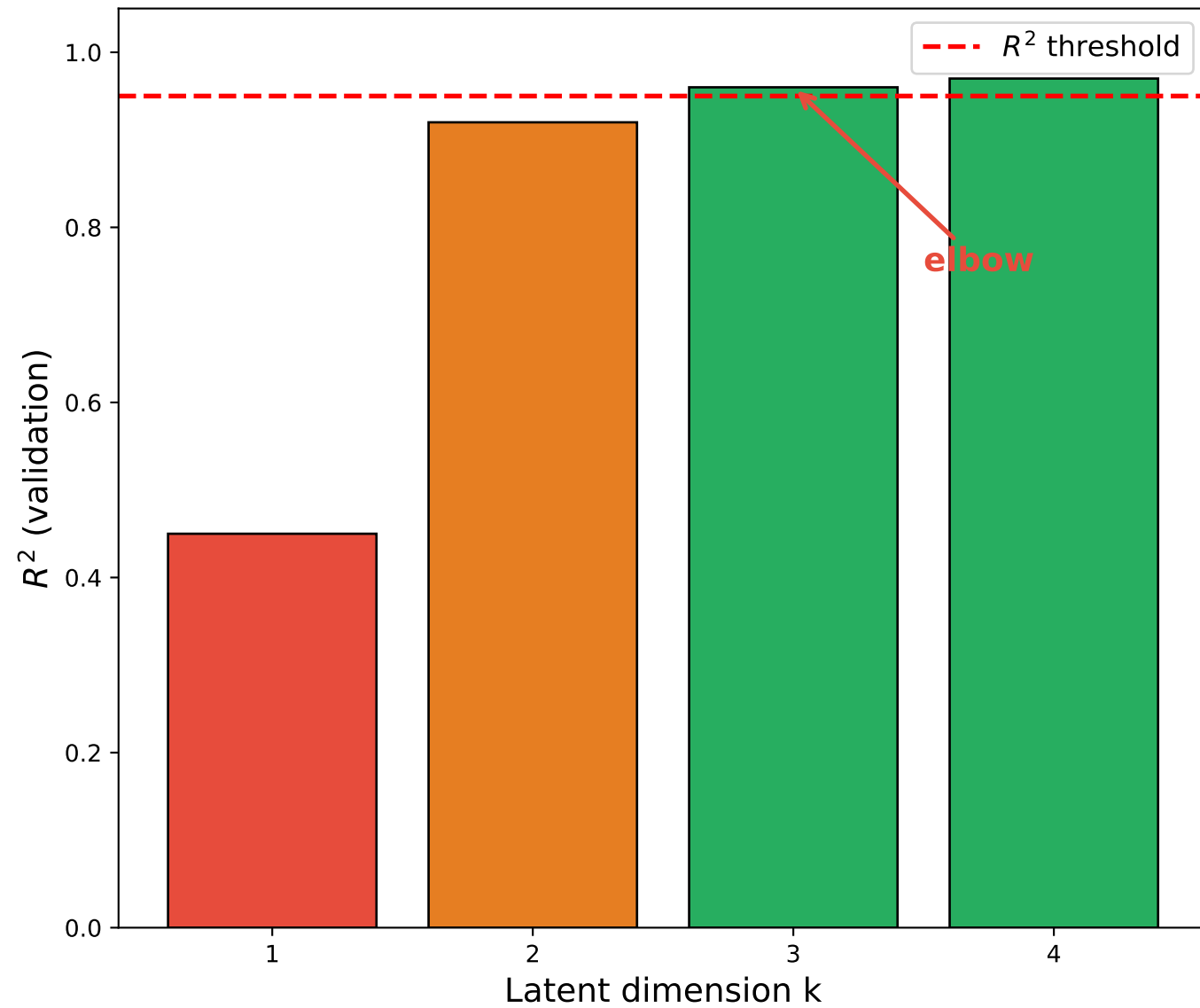
Encoder: single linear layer (default) or multi-layer MLP (optional)
Decoder: always nonlinear 3-layer MLP with Tanh activations

Augmented Feature Space: Why $[X, X^2, \log|X|]$?



Each augmentation lets the linear encoder capture a different symmetry type without prior knowledge

Latent Dimension Selection



Algorithm:

1. Train autoencoder for each k
2. Compute R^2 on validation set
3. Select smallest k where $R^2 > \text{threshold (0.95)}$
4. Report both R^2_{train} and R^2_{test} to check for overfitting

Step 3: Identify Symmetry Type

Translational

$$\phi(x) = x$$

$$z = W \cdot x$$

Rotational

$$\phi(x) = x^2$$

$$z = W \cdot x^2$$

Scaling

$$\phi(x) = \log|x|$$

$$z = W \cdot \log|x|$$

All three share the same frozen decoder from Step 2

Winner = encoder with lowest validation loss

The linear encoder W stays interpretable for generator extraction in Step 4

Step 4: Extract Lie-Algebra Generators

Translational

Generators = null(W)

$$x(\varepsilon) = x_0 + \varepsilon \cdot g$$

Shift along g preserves y

Scaling

Generators = null(W) in log-space

$$x(\varepsilon) = x_0 \cdot e^{\varepsilon \cdot g}$$

Rescaling along g preserves y

Rotational

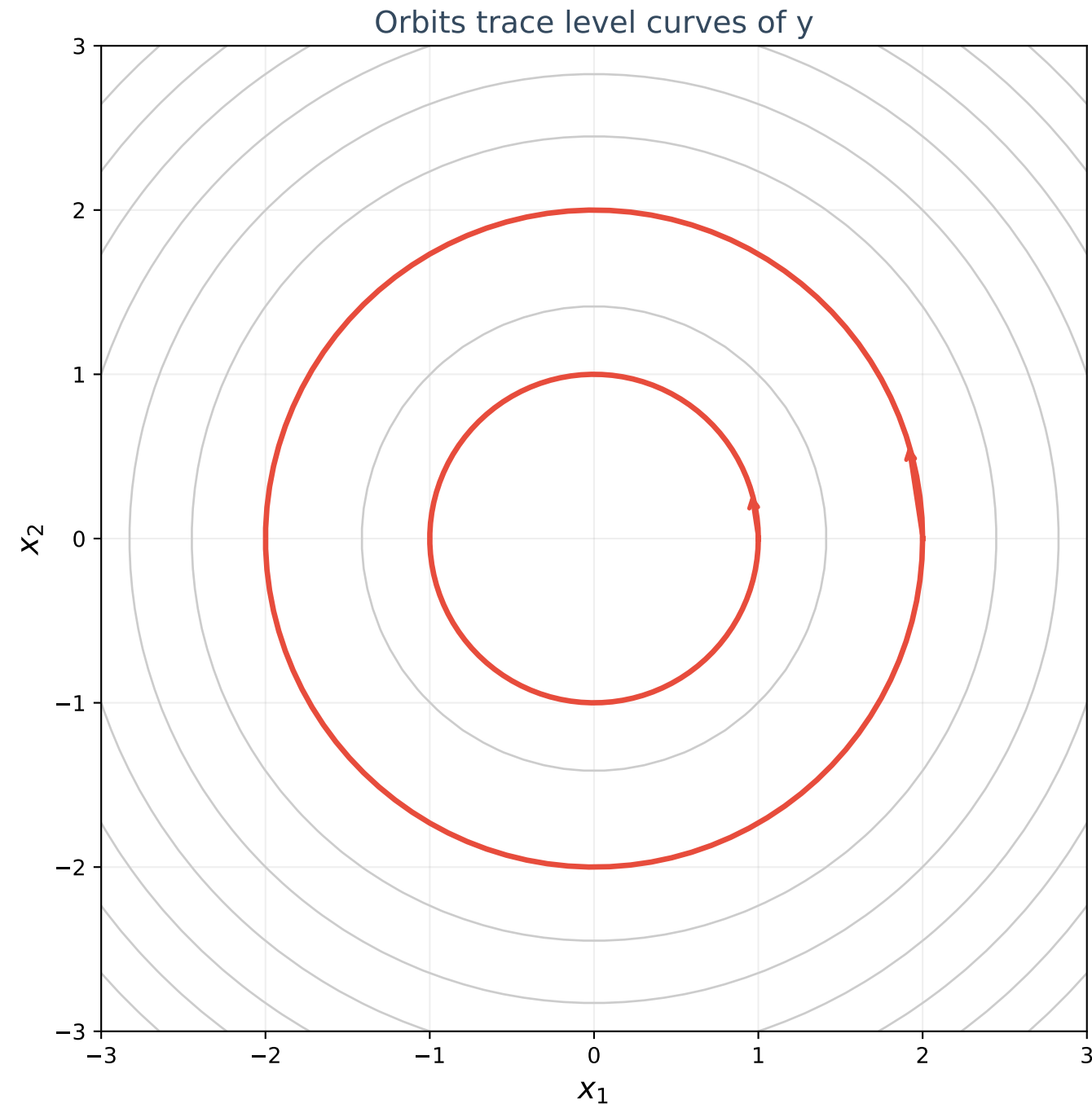
Generators = antisymmetric from W

$$x(\varepsilon) = e^{\varepsilon G} \cdot x_0$$

Rotation by G preserves y

Generators span the Lie algebra of the discovered symmetry group

Generator Orbits: Visualizing Symmetry



Each orbit starts from a data point and traces a path along which the output y stays constant.

Multiple orbits from different starting points trace parallel paths = global symmetry.

Red curves: orbits of a rotational generator

Gray contours: level sets of $y = x_1^2 + x_2^2$

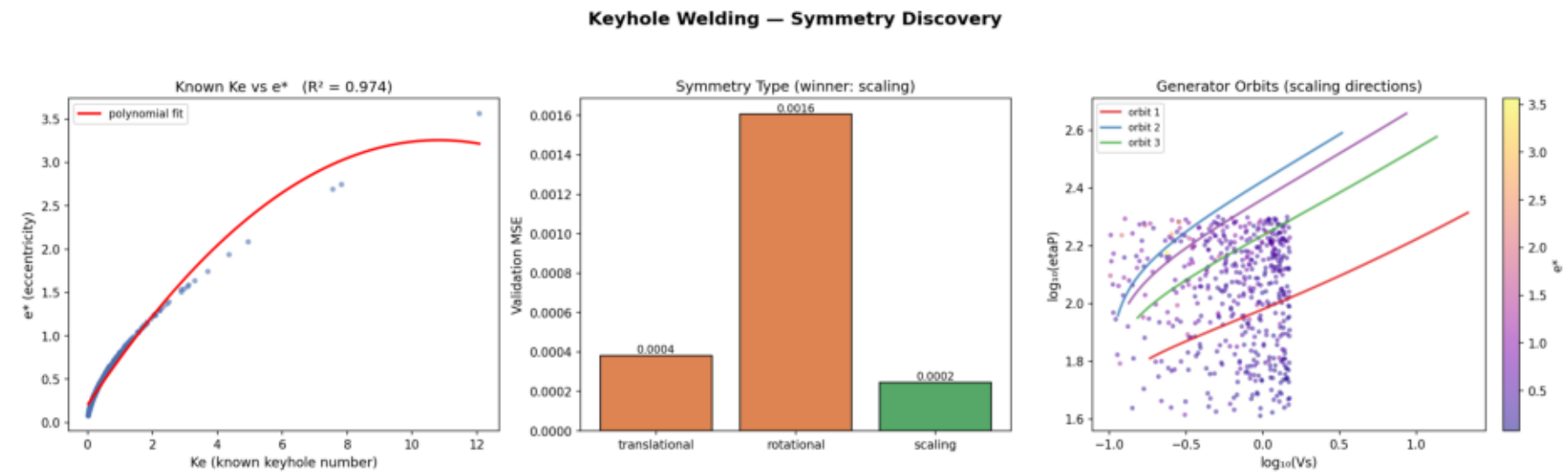
Example: Keyhole Welding — Scaling Symmetry

7 inputs: absorbed power, scan speed,
beam radius, diffusivity, density, cp, temp diff

$$Ke = \frac{\eta P}{\Delta T \cdot \pi \rho c_p \sqrt{\alpha} V_s r_0^3}$$

Detected: Scaling symmetry

Generators: 6 (log-space scaling directions)



Example: Concrete Strength — Translational Symmetry

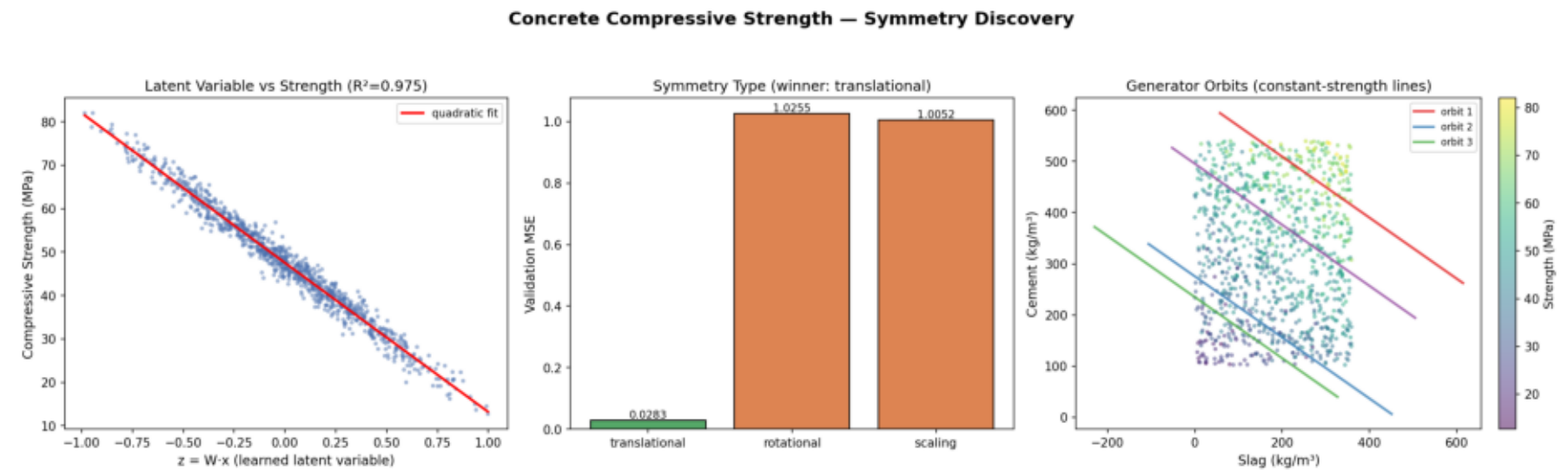
8 inputs: cement, slag, fly ash, water, superplasticizer, coarse/fine aggregate, age

Output: compressive strength (MPa)

Detected: Translational symmetry

Generators: 7 substitution directions

e.g. "increase fly ash, decrease cement"



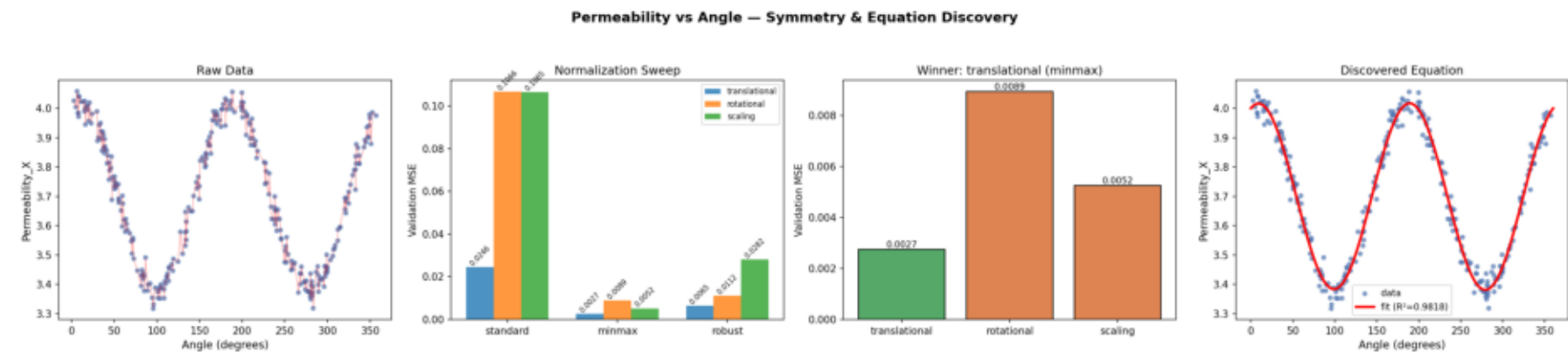
Example: Permeability — Angular Periodicity

180 geometries x 37 angles (0-360 deg)
= 6660 rows

Permeability repeats every 180 degrees

Inputs: $\cos(2\theta)$, $\sin(2\theta)$

Detected: Rotational symmetry
Encodes the 180-degree periodicity



Example: LHC Dijet — Rotational Symmetry

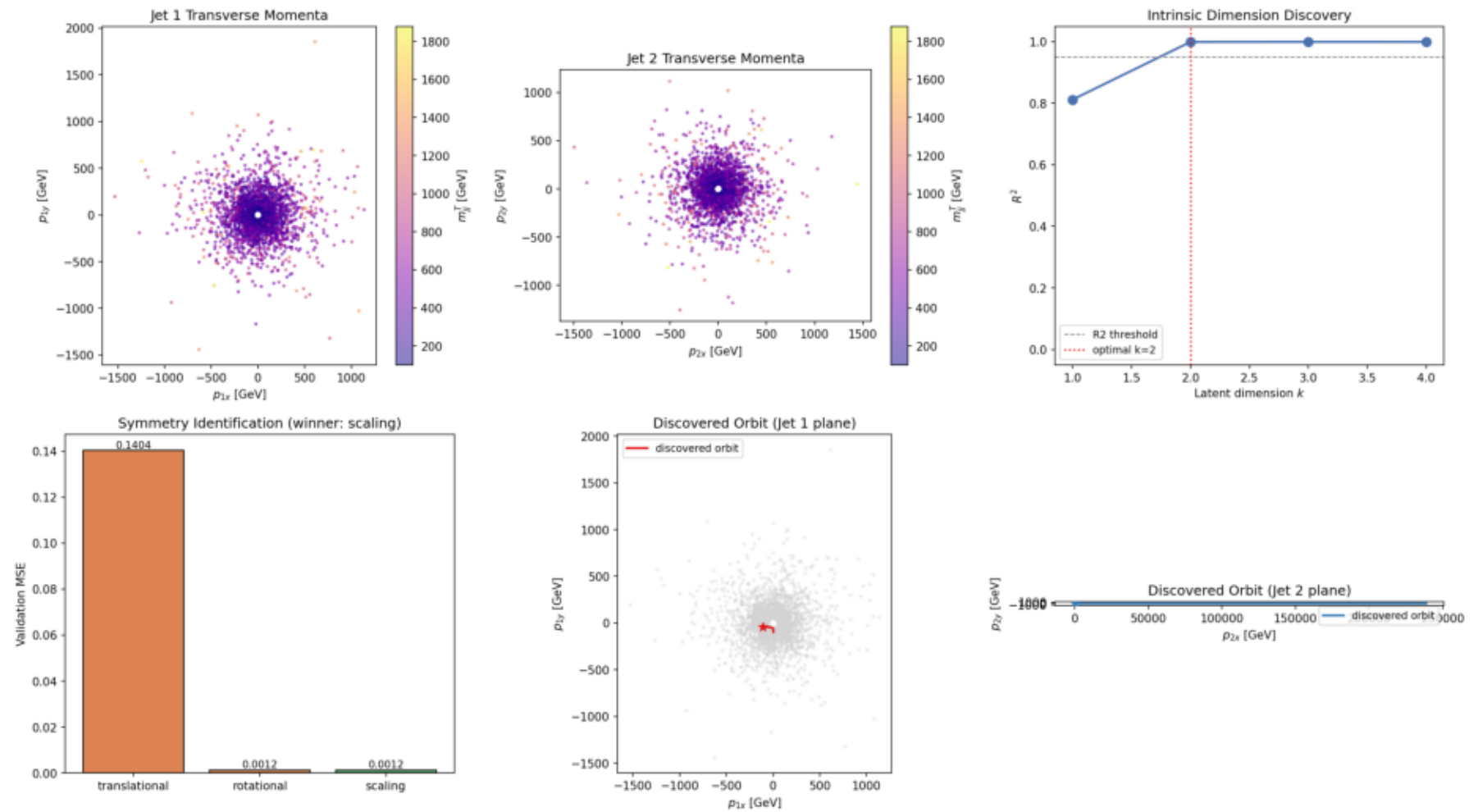
4 inputs: p_{1x} , p_{1y} , p_{2x} , p_{2y}
(anti-kT jet clustering, $R=1.0$)

$$m_{jj}^T = \sqrt{2p_{T1}p_{T2}(1 - \cos \Delta\phi)}$$

Expected: $SO(2)$ azimuthal rotation

Note: y is deterministic from inputs,
so R^2 is artificially high

LHC Dijet Symmetry Discovery



Summary

- Stage1 discovers continuous symmetries from data — no prior physics needed
- 4-step pipeline: Normalize -> Latent Dimension -> Symmetry Type -> Generators
- Three symmetry types: Translational (shifts), Rotational (rotations), Scaling (rescaling)
- Augmented features $[X, X^2, \log|X|]$ let the encoder capture all three types
- Generators reveal the Lie algebra — continuous transformation families preserving physics
- Optional: multi-layer encoder for Step 2, Pi group augmentation for scaling problems
- Validated on: keyhole welding, concrete strength, permeability, LHC dijet data